Construction of a Quarterly Panel of Real GDP per Worker in Constant 2015 USD at Market Exchange Rates

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Abstract

This note documents the construction of an unbalanced country–quarter panel of **real GDP per worker** (employed persons, not population). The series is expressed in a common international unit: constant 2015 US dollars at 2015 period-average market exchange rates. It is not PPP. Exchange rates enter **once**, in the base year, so the time path of each country is the path of real GDP in local currency and is not contaminated by quarterly bilateral-dollar volatility. Employment is official ILOSTAT quarterly data where available, and linearly interpolated official annual data otherwise. The current vintage has 101 countries and 7,450 quarterly observations, 1950Q1–2026Q2. The panel, raw extracts, and code live entirely in this folder; no series from elsewhere in the repository are used.

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1 Purpose and target series

The object of interest is labour productivity at quarterly frequency, for as many economies as national accounts and labour-force data will support:

> annualized real GDP per *employed person*, in constant 2015 USD at 2015 market exchange rates.

Three design constraints follow from that sentence.

1. **Per worker, not per capita.** The denominator is employment (persons), not mid-year population. Using population would mix participation, demography, and productivity.
2. **Common units, not PPP.** Levels must be comparable across countries in the same dollar vintage. Purchasing-power parities are not used. The converter is the 2015 period-average market exchange rate (domestic currency per USD).
3. **No quarterly FX in the volume path.** Converting each quarter with that quarter's exchange rate would put dollar–local-currency swings into measured productivity. Real GDP is therefore taken in local currency at constant prices, turned into a 2015 volume index, scaled by 2015 current-price GDP, and divided by the *2015* average exchange rate only.

The delivered file is `output/gdp_per_worker_q.csv`. Required columns are `country` (ISO 3166-1 alpha-3), `period` (`YYYY-Qn`), `gdp_per_worker`, and `metadata`. Additional columns are audit extras (country name, year, quarter, calendar time, annualized real GDP in 2015 USD, employment in persons, SA/NSA flag, employment source).

2 Construction formula

For country i and quarter t,

$$Y_{i,t}^{\mathrm{USD}} = \left(\frac{Y_{i,t}^{r,q} \times 4}{Y_{i,2015}^{r,A}} \right) \times \left(\frac{Y_{i,2015}^{n,A}}{E_{i,2015}} \right), \qquad \text{gdp per worker}_{i,t} = \frac{Y_{i,t}^{\mathrm{USD}}}{L_{i,t}}. \tag{1}$$

Why 2015. Chain-linked national-accounts volumes have a country-specific reference year (the US QNEA extract is referenced to 2017). Dividing each quarter by that country's 2015 real annual total puts every country on a $2015 = 1$ volume index. Multiplying by 2015 current-price GDP converts the index back into 2015-price local currency. Dividing by the 2015 average FX then puts every country in the same international unit.

Why multiply by 4. IMF QNEA constant-price GDP is a quarterly *flow*, not a seasonally adjusted annual rate. For the United States, 2015Q1 is about 4.67×10^{12}; the four 2015 quarters *sum* to the ANEA 2015 annual total ($\approx 18.80 \times 10^{12}$). Annualizing by 4 makes GDP/employment "output per worker per year."

Table 1: Symbols in (1).

Symbol	Meaning	Source
$Y^{r,q}_{i,t}$	Real GDP, domestic currency, **quarterly flow** (not SAAR)	IMF QNEA `B1GQ`, constant prices, `XDC`
$Y^{r,A}_{i,2015}$	Real GDP, domestic currency, 2015 annual	IMF ANEA `B1GQ` constant `XDC`; fallback = sum of four 2015 QNEA quarters
$Y^{n,A}_{i,2015}$	Current-price GDP, domestic currency, 2015 annual	IMF ANEA `B1GQ` current `XDC`
$E_{i,2015}$	2015 period-average domestic currency per USD	IMF ER `XDC_USD` / `PA_RT` / annual
$L_{i,t}$	Employed persons	ILOSTAT (Section 6)
$\times 4$	Quarterly flow $\rightarrow$ annualized flow	Unit conversion, not seasonal adjustment

Identities (tested). If the four 2015 real quarters sum to the 2015 annual real total, then 2015 annualized Y^{USD} equals 2015 current-price GDP divided by $E_{i,2015}$. Doubling real LCU doubles Y^{USD}. Doubling the base-year FX halves Y^{USD}. The shipped rebase function has no quarterly-FX argument.

United States, 2015. ANEA 2015 current-price GDP is \$18.295 trillion; $E_{\mathrm{US},2015} = 1$. ILO official quarterly employment is about 147–150 million persons. The constructed series is therefore about \$122,500–123,500 per worker in each 2015 quarter, which is the annualized identity above.

3 Sources and exact extracts

Extracts are cached under `raw/` by `download_sources.py`. Re-run with `-force` to refresh. All IMF queries use SDMX 3.0, `Accept: text/csv`. The wildcard `*` is all countries.

3.1 IMF National Economic Accounts

Quarterly (QNEA). Dataset: https://data.imf.org/en/datasets/IMF.STA:QNEA. Key order: `COUNTRY.INDICATOR.PRICE_TYPE.S_ADJUSTMENT.TYPE_OF_TRANSFORMATION.FREQUENCY`. Indicator `B1GQ` is GDP. `Q` in the price slot is constant prices; `XDC` is domestic currency; the last `Q` is quarterly frequency.

- SA volumes: https://api.imf.org/external/sdmx/3.0/data/dataflow/IMF.STA/QNEA/~/*.B1GQ.Q.SA.XDC.Q
- NSA volumes: https://api.imf.org/external/sdmx/3.0/data/dataflow/IMF.STA/QNEA/~/*.B1GQ.Q.NSA.XDC.Q

Annual (ANEA). Dataset: https://data.imf.org/en/datasets/IMF.STA:ANEA. Key order: COUNTRY.INDICATOR.PRICE_TYPE.TYPE_OF_TRANSFORMATION.FREQUENCY.

- Constant-price GDP, LCU, annual: https://api.imf.org/external/sdmx/3.0/data/dataflow/IMF.STA/ANEA/~/*.B1GQ.Q.XDC.A
- Current-price GDP, LCU, annual: https://api.imf.org/external/sdmx/3.0/data/dataflow/IMF.STA/ANEA/~/*.B1GQ.V.XDC.A

IMF SCALE is a power-of-ten unit multiplier ($v \times 10^{\text{SCALE}}$). The GDP extracts used here have SCALE=0 (units of 1 domestic-currency unit). Missing scale is treated as 0. Staff projections and imputations (DERIVATION_TYPE in $\{$SP, SE, SEME, SEHI$\}$) are dropped.

ANEA TIME_PERIOD arrives as float (2015.0) when pandas reads the CSV. The parser accepts 2015, 2015.0, and integer 2015, and rejects nan.

3.2 IMF Exchange Rates

Dataset: https://data.imf.org/en/datasets/IMF.STA:ER. Key order: COUNTRY.INDICATOR.TYPE_OF_TRANSFORMATION.FREQUENCY. XDC_USD is domestic currency per US dollar; PA_RT is period average; frequency is annual.

https://api.imf.org/external/sdmx/3.0/data/dataflow/IMF.STA/ER/~/*.XDC_USD.PA_RT.A

Country names come from IMF codelist CL_COUNTRY: https://api.imf.org/external/sdmx/3.0/structure/codelist/IMF/CL_COUNTRY.

3.3 ILOSTAT employment

Bulk CSVs via the ILO Rplumber indicator service (https://ilostat.ilo.org/data/):

- Official, quarterly: https://rplumber.ilo.org/data/indicator/?id=EMP_TEMP_SEX_AGE_NB_Q
- Official, annual: https://rplumber.ilo.org/data/indicator/?id=EMP_TEMP_SEX_AGE_NB_A
- ILO modelled estimates, annual (downloaded as a last-resort source; not used in the current vintage, because every panel country had official data): https://rplumber.ilo.org/data/indicator/?id=EMP_2EMP_SEX_AGE_NB_A

ILO time labels are YYYYQn (no hyphen). IMF labels are YYYY-Qn. Both parse to the same year and quarter.

4 Seasonal adjustment of GDP

QNEA publishes both SA and NSA. IMF SA and NSA GDP are never spliced inside a country:

- If a country has any SA constant-price GDP, all of that country's GDP observations come from SA.
- Otherwise GDP is NSA. Productivity is then seasonally adjusted with statsmodels SARIMAX when the spell is long enough.

SARIMAX on NSA countries. NSA GDP, and typically NSA employment, leave seasonality in the productivity ratio. For those countries, if there are at least 12 observations covering all four calendar quarters, we fit

$$\log(\text{gdp per worker})_t = a + gt + \gamma_{q(t)} + u_t, \qquad u_t \sim \text{ARMA}(p, q), \tag{2}$$

as `SARIMAX(log` y`, order=(`p`,0,`q`), trend=ct, exog=Q2-Q4 dummies)`. (p, q) is chosen by AIC. `d=0` so dummy coefficients are *level* seasonal factors (differencing would turn dummies into pulses). The four γs are recentered to sum to zero. Dummies use the calendar quarter, not the row position. After adjustment, `gdp_real_2015usd` is rebuilt as `gdp_per_worker` $\times$ `emp_persons`, and `gdp_sa` is set to `SARIMAX`.

Spells shorter than 12 quarters, or missing a calendar quarter, stay `NSA`. In the current vintage that is Ghana (11), El Salvador (10), Belarus (6), Bangladesh (4), Botswana (2), and Mozambique (2).

The current vintage: 68 IMF-SA, 27 SARIMAX, 6 remaining NSA.

5 Exchange rates, including the euro

$E_{i,2015}$ is the 2015 annual average of domestic currency per USD. For the United States the value is 1. For the United Kingdom it is the 2015 average pound per dollar.

Euro-area members as of 1 January 2015—Austria, Belgium, Cyprus, Germany, Spain, Estonia, Finland, France, Greece, Ireland, Italy, Lithuania, Luxembourg, Latvia, Malta, the Netherlands, Portugal, Slovakia, Slovenia—have national `XDC_USD` series that *stop at euro adoption* (Germany: 1998). After adoption the domestic currency is the euro. Those countries, and euro users without a 2015 national series (Andorra, Kosovo, Montenegro, San Marino, Vatican), inherit the 2015 Euro Area aggregate rate, IMF country code `G163` ($\approx$ 0.901 EUR per USD in 2015).

That inherited rate is still a single 2015 converter. Croatia is not in the set: in 2015 it used the kuna, so its own 2015 `XDC_USD` series is the correct converter. `G163` is an FX donor only; it is never a panel country.

6 Employment

The denominator is employed persons, both sexes. ILO `EMP_*_NB` values are thousands of persons and are multiplied by 1,000.

6.1 Filters

Applied to every ILO extract:

- `sex = SEX_T` (both sexes). Male/female breakdowns are discarded.
- Age preference, one definition per country, never mixed:
  1. `AGE_YTHADULT_YGE15` (15+)
  2. `AGE_AGGREGATE_YGE15`
  3. `AGE_AGGREGATE_TOTAL`
  4. `AGE_YTHADULT_Y15-64`
- One ILO `source` code per country (the longest series), so LFS vintages are not spliced.

6.2 Source hierarchy (country-level)

1. Official **quarterly** `EMP_TEMP_SEX_AGE_NB_Q`, if the country has at least eight quarterly observations after filters.
2. Otherwise official **annual** `EMP_TEMP_SEX_AGE_NB_A`, linearly interpolated to quarter midpoints.
3. Otherwise ILO modelled annual `EMP_2EMP_SEX_AGE_NB_A`, interpolated the same way.

Sources are not spliced quarter-by-quarter inside a country, except for the narrow $4\times$ patch in Section 7. In the current vintage, 86 countries use official quarterly employment and 15 use interpolated official annual. No country falls through to modelled estimates.

6.3 Annual to quarterly interpolation

The annual figure for year y is treated as the annual *average* stock, located at mid-year $t = y + 0.5$. Quarter q of year y is located at

$$t = y + (q - 0.5)/4 \qquad \text{(Q1: } y + 0.125,\ \ldots,\ \text{Q4: } y + 0.875\text{)}. \tag{3}$$

Between adjacent annual observations the interpolant is the unique straight line in t. Quarters whose midpoint lies outside $[t_{\text{first}}, t_{\text{last}}]$ are dropped (no extrapolation).

This is interpolation of a stock, not Denton benchmarking. The four-quarter average equals the annual level only if employment is linear in calendar time. That limitation is accepted in exchange for a transparent, testable rule.

7 Data-quality repairs

Raw QNEA and ILO extracts contain a small number of unit and survey breaks that would otherwise dominate the productivity series. The repairs below are implemented in `construct.py` and covered by specification tests.

7.1 Power-of-ten unit breaks in QNEA

Some QNEA LCU series change unit (1 $\rightarrow$ thousands $\rightarrow$ millions) without a matching `SCALE`. Jamaica in 2018 is a million-fold drop in the raw extract while ANEA annual real GDP does not drop.

For each country-year compare $4 \times \text{mean}(\text{quarterly LCU})$ to ANEA annual real LCU. If they differ by 10^k with $|k| \geq 3$, that year's quarterly LCU is multiplied by 10^{-k}. A SAAR-versus-flow gap of 4 ($\log_{10} 4 \approx 0.6$) is ignored. Years without ANEA inherit the neighbouring year's factor. The factor is recorded in `metadata` when it is not 1.

7.2 Years that still disagree with ANEA

After that rescale, country-years whose $4 \times \text{mean}(\text{quarterly})$ still differs from ANEA annual real by more than a factor of 5 are **dropped**. Residual breaks cannot be repaired with a power of ten without putting FX back into the volume path.

7.3 Spells that do not contain the rebase year

A remaining adjacent-quarter real-GDP move larger than a factor of 20 is treated as a unit or currency restatement (El Salvador's colon/USD overlap around dollarization). Only the contiguous spell that contains 2015 is kept, so the volume index is never a ratio of two different currency units. El Salvador in the panel therefore starts in 2005Q1 rather than 2000Q1.

7.4 ILO quarterly employment off the annual path

Official quarterly employment is occasionally a factor of about four too small for a calendar year. Israel 2017 is the textbook case: about 0.94 million persons in the quarterly extract against about 3.7 million on the annual path.

Where an official annual series exists, a quarter whose employment divided by the interpolated annual path lies outside [0.4, 2.5] is replaced by that interpolant and the source string is annotated. Israel 2017 in the panel is about 3.81 million persons, in line with 2016 and 2018.

8 Sample and join

Keep ISO 3166-1 alpha-3 codes (and Kosovo `KOS` if present). Drop IMF aggregates. The panel is the **inner join** of constructed GDP and employment on (country, period). Duplicate country–period rows are rejected. Non-positive GDP or employment is dropped.

The binding constraint is IMF quarterly real GDP, not employment. QNEA constant-price LCU exists for roughly a hundred economies; the inner join with 2015 annual levels, 2015 FX, and ILO employment yields 101 countries.

9 Output schema and metadata

Table 2: Columns in output/gdp_per_worker_q.csv.

Column	Required?	Content
country	yes	ISO3
period	yes	YYYY-Qn
gdp_per_worker	yes	annualized 2015 USD per employed person
metadata	yes	sources and transformations for the observation
country_name	no	IMF CL_COUNTRY label
year, quarter, time	no	calendar parts; time$= y + (q-1)/4$
gdp_real_2015usd	no	annualized real GDP, 2015 USD
emp_persons	no	employed persons
gdp_sa	no	SA or NSA
emp_source	no	ILO series and any interpolation/patch note

Each observation's metadata records, in order: the QNEA series and SA/NSA flag and the $\times 4$ annualization (plus any power-of-ten rescale); the 2015 real and nominal level sources; the 2015 FX source (including G163 when used); the ILO series, source code, sex, age, and thousands-to-persons conversion; and that the ratio is 2015 USD market FX, not PPP.

10 Replication

From this folder:

```
python download_sources.py
python build_gdp_per_worker.py
python -m unittest tests.test_construct
```

construct.build_panel(...) is the construction entry point. The CLI only reads the cached CSVs into that function. Tests call the shipped functions (including build_panel) and check documented identities: the 2015 USD level, one-for-one pass-through of real LCU growth, euro FX fill from G163, SA-not-mixed-with-NSA, mid-year interpolation geometry, no extrapolation, power-of-ten unit alignment, $4\times$ employment patches, and dropping of IMF staff projections.

11 Current vintage

Coverage by country is Table 4. Longest samples include the United States (1950Q1–2026Q2, $N = 306$), Korea ($N = 242$), Canada ($N = 202$), and Australia ($N = 192$). Several low-income economies appear with short spells (Bangladesh 2024 only; Botswana and Mozambique two quarters). Those short spells are retained: the product is an unbalanced panel, and the estimator that consumes it already drops countries below a minimum T.

Table 3: Headline counts, vintage of the raw/ extracts dated 10 September 2026.

Countries	101
Observations	7,450
Span	1950Q1–2026Q2 (unbalanced)
GDP SA / SARIMAX / NSA (country-level)	68 / 27 / 6
Employment: official quarterly / interpolated annual	86 / 15
United States, 2015 mean GDP per worker	$122,925
Germany, 2015 mean GDP per worker	$85,222

12 Code map

File	Role
download_sources.py	URLs, streaming download, raw/ cache
construct.py	Every transformation listed above
build_gdp_per_worker.py	CLI: download if needed → build → output/
tests/test_construct.py	Specification tests against shipped functions
docs/gdp_per_worker_construction.tex	This note

13 Citations

- IMF, National Economic Accounts, Quarterly (QNEA) and Annual (ANEA).
- IMF, Exchange Rates (ER), domestic currency per US dollar, period average.
- ILOSTAT, Employment by sex and age (EMP_TEMP_SEX_AGE_NB); ILO modelled estimates (EMP_2EMP_SEX_AGE_NB) as a last-resort source (unused in this vintage).

Cite the vintage of the cached files in raw/ (download date = file timestamps).

A Country coverage

Table 4: Country coverage of the constructed panel. Emp. is the country-level employment source: official quarterly ILO (Q official); official quarterly with isolated 4× patches from interpolated annual (Q, patched); or linearly interpolated official annual (A interpolated). SA/NSA is the country-level GDP seasonal-adjustment choice.

ISO3	Country	SA	N	Start	End	Emp.
USA	United States	SA	306	1950-Q1	2026-Q2	Q official
KOR	Korea, Republic of	SA	242	1966-Q1	2026-Q2	Q official
CAN	Canada	SA	202	1976-Q1	2026-Q2	Q official
AUS	Australia	SA	192	1978-Q2	2026-Q1	Q official
NZL	New Zealand	SA	156	1987-Q2	2026-Q1	Q official
NOR	Norway	SA	139	1990-Q1	2026-Q1	Q official
SWE	Sweden	SA	133	1993-Q1	2026-Q1	Q official
JPN	Japan	SA	130	1994-Q1	2026-Q2	Q official
SVK	Slovak Republic	SA	125	1995-Q1	2026-Q1	Q official
ESP	Spain	SA	125	1995-Q1	2026-Q1	Q official
CZE	Czech Republic	SA	125	1995-Q1	2026-Q1	Q official
HUN	Hungary	SA	122	1995-Q1	2026-Q1	Q official
PRT	Portugal	SA	121	1995-Q2	2026-Q1	Q official
ISR	Israel	SA	120	1995-Q1	2024-Q4	Q official
ITA	Italy	SA	117	1996-Q2	2026-Q1	Q official
FIN	Finland	SA	116	1995-Q2	2026-Q1	Q official
GRC	Greece	SA	116	1995-Q2	2026-Q1	Q official
SVN	Slovenia, Republic of	SA	114	1996-Q2	2026-Q1	Q official
BEL	Belgium	SA	113	1995-Q2	2026-Q1	Q official
FRA	France	SA	113	1983-Q1	2026-Q1	Q official
AUT	Austria	SA	113	1995-Q1	2026-Q1	Q official
DNK	Denmark	SA	113	1995-Q2	2026-Q1	Q official
IRL	Ireland	SA	112	1995-Q2	2026-Q1	Q official
ROU	Romania	SA	111	1997-Q2	2026-Q1	Q official
MEX	Mexico	SA	110	1995-Q2	2026-Q2	Q official
LVA	Latvia, Republic of	SA	109	1996-Q2	2026-Q1	Q official
NLD	Netherlands, The	SA	109	1996-Q2	2026-Q1	Q official
EST	Estonia, Republic of	SA	108	1997-Q2	2026-Q1	Q official
BGR	Bulgaria	SA	105	2000-Q1	2026-Q1	Q official
LTU	Lithuania, Republic of	SA	105	1998-Q2	2026-Q1	Q official
POL	Poland, Republic of	SA	105	2000-Q1	2026-Q1	Q official
HRV	Croatia, Republic of	SA	105	1998-Q1	2026-Q1	Q official
MLT	Malta	SA	103	2000-Q2	2026-Q1	Q official
HND	Honduras	SA	102	2000-Q1	2025-Q2	A interpolated
LUX	Luxembourg	SA	101	1995-Q2	2026-Q1	Q official
ISL	Iceland	SA	100	1995-Q2	2025-Q4	Q official
PHL	Philippines	SA	99	2000-Q1	2024-Q4	Q official
DEU	Germany	SA	95	1991-Q2	2026-Q1	Q official
CYP	Cyprus	SA	93	1999-Q2	2026-Q1	Q official
MKD	North Macedonia, Republic of	SA	89	2004-Q1	2026-Q1	Q official
ZAF	South Africa	SA	86	2000-Q1	2025-Q2	Q official
GBR	United Kingdom	SA	85	2005-Q1	2026-Q1	Q official
ARG	Argentina	SA	84	2004-Q1	2025-Q4	Q official
TUR	Türkiye, Republic of	SA	81	2006-Q1	2026-Q1	Q official
JAM	Jamaica	SA	80	2000-Q1	2023-Q4	Q official
CHE	Switzerland	SA	79	1996-Q2	2026-Q1	Q official
SGP	Singapore	SA	78	2000-Q1	2025-Q4	Q official
COL	Colombia	SA	77	2007-Q1	2026-Q1	Q official

Continued on next page

ISO3	Country	SA	N	Start	End	Emp.
LSO	Lesotho, Kingdom of	SA	70	2007-Q1	2024-Q2	A interpolated
KAZ	Kazakhstan, Republic of	SA	68	2003-Q1	2019-Q4	Q official
CHL	Chile	SA	65	2010-Q1	2026-Q2	Q official
ECU	Ecuador	SA	64	2003-Q4	2025-Q3	Q official
CRI	Costa Rica	SA	63	2010-Q3	2026-Q1	Q official
SRB	Serbia, Republic of	SA	60	2008-Q2	2025-Q4	Q official
MNE	Montenegro	SA	60	2010-Q1	2024-Q4	Q official
RUS	Russian Federation	SARIMAX	60	2011-Q1	2025-Q4	Q official
LKA	Sri Lanka	SARIMAX	58	2010-Q1	2024-Q4	Q official
THA	Thailand	SA	57	2010-Q1	2026-Q1	Q official
BRA	Brazil	SA	57	2012-Q1	2026-Q1	Q official
ALB	Albania	SA	52	2012-Q1	2024-Q4	Q official
TTO	Trinidad and Tobago	SARIMAX	46	2012-Q1	2024-Q4	Q official
MDV	Maldives	SARIMAX	46	2003-Q1	2014-Q2	A interpolated
DOM	Dominican Republic	SA	45	2015-Q1	2026-Q1	Q official
MUS	Mauritius	SARIMAX	45	2013-Q1	2024-Q4	Q official
PER	Peru	SARIMAX	44	2011-Q1	2021-Q4	Q official
BRN	Brunei Darussalam	SARIMAX	44	2014-Q3	2025-Q2	A interpolated
MNG	Mongolia	SARIMAX	44	2015-Q1	2025-Q4	Q official
UKR	Ukraine	SA	42	2010-Q1	2020-Q2	Q official
CPV	Cabo Verde	SARIMAX	40	2009-Q3	2019-Q2	A interpolated
GMB	Gambia, The	SARIMAX	40	2014-Q1	2023-Q4	A interpolated
KOS	Kosovo, Republic of	SARIMAX	40	2012-Q1	2021-Q4	Q official
BHS	Bahamas, The	SARIMAX	38	2015-Q1	2024-Q2	A interpolated
PRY	Paraguay	SARIMAX	36	2017-Q1	2025-Q4	Q official
IND	India	SA	34	2017-Q3	2025-Q4	Q official
IDN	Indonesia	SA	32	2008-Q1	2023-Q3	Q official
NIC	Nicaragua	SARIMAX	30	2006-Q1	2013-Q2	A interpolated
RWA	Rwanda	SARIMAX	28	2017-Q1	2025-Q4	Q official
LCA	St. Lucia	SARIMAX	28	2017-Q1	2024-Q3	Q official
ARM	Armenia, Republic of	SARIMAX	24	2008-Q1	2013-Q4	Q official
SYC	Seychelles	SARIMAX	23	2016-Q1	2024-Q2	Q, patched
NGA	Nigeria	SARIMAX	23	2014-Q1	2024-Q2	Q official
SEN	Senegal	SA	23	2016-Q2	2025-Q1	Q official
BIH	Bosnia and Herzegovina	SA	23	2020-Q1	2025-Q3	Q official
GEO	Georgia	SARIMAX	23	2020-Q1	2025-Q3	Q official
NAM	Namibia	SARIMAX	22	2013-Q1	2018-Q2	A interpolated
EGY	Egypt, Arab Republic of	SARIMAX	21	2019-Q4	2024-Q4	Q official
ZMB	Zambia	SARIMAX	20	2017-Q1	2024-Q4	Q official
GTM	Guatemala	SARIMAX	20	2013-Q2	2025-Q3	Q official
CHN	China, People’s Republic of	SARIMAX	18	2011-Q1	2015-Q2	A interpolated
QAT	Qatar	SARIMAX	16	2013-Q1	2016-Q4	Q official
PAK	Pakistan	SARIMAX	16	2017-Q3	2025-Q4	Q official
MAR	Morocco	SA	15	2014-Q1	2017-Q3	Q official
UGA	Uganda	SA	12	2016-Q3	2019-Q2	A interpolated
GHA	Ghana	NSA	11	2022-Q1	2024-Q3	Q official
SLV	El Salvador	NSA	10	2005-Q1	2007-Q2	A interpolated
SAU	Saudi Arabia	SA	8	2012-Q1	2016-Q1	Q official
KEN	Kenya	SA	8	2019-Q1	2021-Q4	Q official
BLR	Belarus, Republic of	NSA	6	2014-Q1	2015-Q2	A interpolated
BGD	Bangladesh	NSA	4	2024-Q1	2024-Q4	Q official
BWA	Botswana	NSA	2	2006-Q1	2006-Q2	A interpolated
MOZ	Mozambique, Republic of	NSA	2	2022-Q1	2022-Q2	A interpolated